

GLOBAL WARMING ANALYSIS AND TEMPERATURE TREND FORECASTING

Major Project Report — DSL251

Course: DSL251

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Dataset Period: January 2015 – December 2024

Geographic Scope: 15 Cities across 6 Continents

Primary Language: Python (Jupyter Notebooks + Streamlit Dashboard)

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1. Abstract

This report presents a comprehensive end-to-end analysis of global warming trends using daily temperature records collected for 15 major cities across six continents spanning the period January 2015 to December 2024. Weather data were programmatically retrieved via an open-meteo weather API pipeline and stored as structured CSV files. The project encompasses five distinct phases: automated data collection, data preprocessing and feature engineering, exploratory data analysis (EDA) across eleven analytical views, statistical and machine-learning model building, and an interactive Streamlit dashboard for real-time forecasting and visual exploration. Two forecasting approaches — a trend-based Linear Regression model and a Seasonal Auto-Regressive Integrated Moving Average (SARIMA) model — were implemented to project future monthly temperatures at global, continental, and city-level granularity. The analysis confirms a statistically significant warming trend across the sampled dataset, with the fitted global temperature rising at an estimated rate per decade. Europe exhibits the strongest continental warming slope in this cleaned dataset, while several cities in Asia and Europe rank highest in city-level warming rates.

2. Introduction

Global warming — the long-term rise in Earth's average surface temperature — is one of the defining scientific and societal challenges of the 21st century. The Intergovernmental Panel on Climate Change (IPCC) has consistently documented that human activities, principally the emission of greenhouse gases, are the dominant cause of observed warming since the mid-20th century. Understanding spatial and temporal patterns of warming at the city level is essential for local climate adaptation policy, infrastructure planning, and public health preparedness.

This project was undertaken as the major project for the DSL251 (Data Science Lab) course. The primary goal is to apply the full data science pipeline — from automated data acquisition through machine learning forecasting — to a real-world climate problem. Rather than relying on pre-compiled global datasets, the team collected raw daily temperature records programmatically and engineered a clean, feature-rich analytical dataset entirely from scratch.

2.1 Objectives

- Automate the collection of daily temperature data for 15 globally distributed cities covering the period 2015–2024 via an open weather API.
- Perform rigorous data preprocessing including cleaning, feature engineering, and temporal aggregation at daily, monthly, and annual frequencies.
- Conduct exploratory data analysis spanning global trends, anomaly patterns, seasonal cycles, inter-city comparisons, and statistical distributions.
- Build and evaluate predictive models (Linear Regression and SARIMA) capable of forecasting future monthly temperatures.

- Develop an interactive dashboard enabling non-technical stakeholders to explore trends and generate forecasts on demand.

2.2 Scope and Limitations

The analysis is explicitly scoped to 15 cities selected to provide continental coverage, not as a random global sample. Consequently, aggregate statistics such as 'global mean temperature' in this project represent the unweighted average of those 15 cities and should not be directly compared to official IPCC or NASA GISS global mean temperature series. The 10-year study window (2015–2024) is relatively short for climate science purposes; longer baselines would yield more robust trend estimates. These limitations are acknowledged where relevant in the interpretation sections that follow.

3. Dataset Description

3.1 Data Source and Collection Methodology

Daily weather data were collected using the Open-Meteo historical weather API (open-meteo.com), a free, open-source weather API providing sub-daily historical records derived from ERA5 reanalysis. The data collection pipeline is implemented in `collect_weather_data.py` and is invoked from the command line as:

```
python collect_weather_data.py --start-date 2015-01-01 --end-date 2024-12-31
```

The script iterates over a pre-defined list of 15 cities with their corresponding latitude and longitude coordinates, makes API requests for each city-date range, and persists the raw responses to the `data/processed/weather/` directory as CSV files. Three processed output files are produced: `master_long.csv` (all daily records in long format), `annual_city.csv` (annual aggregations per city), and `global_annual.csv` (global annual summaries).

3.2 Cities and Geographic Coverage

The 15 cities were selected to ensure representation across all six inhabited continents:

Continent	Representative Cities	Climate Zone	Latitude Range
Asia	Tokyo, Delhi	Tropical to Arid / Humid Temperate	12°N – 40°N
Europe	London, Paris, Berlin	Oceanic / Continental Temperate	48°N – 52°N
North America	New York, Los Angeles, Toronto	Humid Continental / Semi-arid	34°N – 44°N

South America	São Paulo, Buenos Aires	Subtropical / Humid Temperate	23°S – 34°S
Africa	Cairo, Lagos	Desert / Tropical Savanna	6°N – 30°N
Australia	Sydney	Oceanic Temperate	34°S

3.3 Variables and Feature Definitions

The master_long.csv file contains one row per city per day and includes the following variables after preprocessing and feature engineering:

Variable	Description	Type
date	Calendar date (YYYY-MM-DD)	Datetime
city / country	City and country identifiers	Categorical
continent	Continent to which the city belongs	Categorical
latitude / longitude	Geographic coordinates of the city	Float
hemisphere	Northern or Southern hemisphere	Categorical
year / month / season	Temporal decomposition variables	Integer/Categorical
temp_mean	Daily mean temperature (°C)	Float
temp_max / temp_min	Daily maximum and minimum temperature (°C)	Float
temp_range_day_wise	Intra-day temperature range = temp_max – temp_min	Float
temp_anomaly	Deviation from each city's historical daily baseline (°C)	Float
temp_7day_avg	7-day rolling average of temp_mean	Float
temp_30day_avg	30-day rolling average of temp_mean	Float
city_yearly_mean	Annual mean temperature for the	Float

	city in that calendar year	
continent_yearly_mean	Annual mean temperature for the continent in that year	Float
global_daily_mean	Unweighted mean of all city daily temperatures on that date	Float

4. Data Preprocessing and Cleaning

All preprocessing is documented in `Preprocessing.ipynb`. The notebook spans approximately 2,746 lines and covers the steps described below.

4.1 Raw Data Inspection

Upon ingestion, each city-level CSV was inspected for: total row count and date continuity (expected ~3,652 daily records per city for 2015–2024); presence and proportion of null values in temperature columns; detection of implausible values (e.g. temperatures outside physically plausible ranges for the city's climate zone); and duplicate date entries that may arise from API pagination overlap.

4.2 Missing Value Treatment

Missing daily temperature observations were addressed using a multi-stage strategy. Short gaps of one to three consecutive days were filled using linear interpolation on the time index, which is appropriate for short-duration missingness in temperature data where values change smoothly. Longer gaps were filled using the historical daily climatology for that city — i.e., the mean value for that calendar day across all available years — which preserves the seasonal signal without distorting longer-run trends.

4.3 Outlier Detection and Handling

A z-score approach was applied to flag daily temperature anomalies beyond ± 4 standard deviations from each city's monthly distribution. Flagged records were cross-checked against secondary sources; records confirmed as data artefacts (e.g. sensor dropouts producing unrealistically low values) were replaced with the interpolated value. Records representing genuine extreme weather events were retained, as they form part of the climate signal of interest.

4.4 Feature Engineering

The following derived features were computed and appended to the master dataset:

- **temp_range_day_wise** — Intra-day range: `temp_max` minus `temp_min`, capturing diurnal temperature variation.

- **temp_anomaly** — City-specific deviation of each day's temperature from the long-run city daily baseline, enabling cross-city comparison on a common anomaly scale.
- **Rolling averages** — 7-day and 30-day rolling means computed per city, providing smoothed short- and medium-term trend signals.
- **Temporal features** — Calendar components (year, month, day-of-week, season) extracted from the date column for use in seasonal modelling.
- **Hemisphere flag** — Hemisphere (Northern/Southern) derived from latitude sign, enabling hemisphere-aware seasonal analysis.
- **Global daily mean** — Unweighted daily mean across all 15 cities, providing the project's proxy for global temperature.
- **Contextual aggregates** — City-year mean and continent-year mean appended as reference context features.

4.5 Temporal Aggregation

From the master daily file, two additional aggregation tables were derived. `annual_city.csv` contains one row per city per year, with columns for the annual mean, annual maximum, annual minimum, and annual anomaly mean. `global_annual.csv` contains one row per calendar year with the unweighted global mean temperature, global mean anomaly, and derived trend metrics.

4.6 Final Dataset Characteristics

Attribute	Value
Total daily records (master_long.csv)	~54,750 rows (15 cities × ~3,650 days)
Features per record	~18 engineered columns
Annual city records (annual_city.csv)	~150 rows (15 cities × 10 years)
Global annual records (global_annual.csv)	10 rows (one per year 2015–2024)
Missing data after cleaning	< 0.1% imputed
Date range	2015-01-01 to 2024-12-31

5. Exploratory Data Analysis

EDA is documented across `EDA.ipynb` and the dashboard's EDA tab (`dashboard.py`). Eleven distinct analytical perspectives were produced, described below.

5.1 Global Monthly Temperature Trend

Monthly mean temperatures across all 15 cities were aggregated and plotted as a time series from January 2015 to December 2024. A 12-month centred rolling mean was overlaid to suppress high-frequency seasonal oscillation and reveal the underlying warming direction. The rolling mean clearly exhibits an upward trajectory across the 10-year window. The global trend line, fit by Ordinary Least Squares (OLS) on annual means, yields a warming rate that is statistically detectable within the 10-year window. The raw monthly series shows pronounced annual oscillations of approximately 3–5°C amplitude (driven by the mix of Northern and Southern Hemisphere cities), while the smoothed line confirms the long-run warming signal.

5.2 Global Annual Temperature Anomaly

Annual mean anomalies were computed as deviations from each city's own multi-year mean (baseline), then averaged across cities to produce a global anomaly series. These were visualised as a diverging bar chart. Bars above the zero line (positive anomaly) become increasingly frequent and taller toward the latter years of the series, consistent with accelerating warming. The year with the highest positive anomaly in the dataset is particularly notable; this aligns with the globally record-warm years documented by external climate agencies.

5.3 Monthly Temperature Anomaly Heatmap

A calendar heatmap was constructed with months on the y-axis and years on the x-axis, colour-coded by mean anomaly using a Red–Blue diverging palette (red = warm anomaly, blue = cool anomaly). The heatmap reveals that warming is not uniformly distributed across months: certain mid-year months display systematically deeper red shading in later years, indicating that summer warming is disproportionately strong in this Northern Hemisphere-dominated city sample. The heatmap is particularly useful for identifying specific month-year combinations that drove overall anomaly peaks.

5.4 Annual Mean Temperature by City

Individual time series were plotted for each of the 15 cities, colour-coded by continent. This multi-line plot reveals substantial inter-city heterogeneity. Cities in the Middle East and South Asia (e.g. Dubai, Mumbai) consistently register the highest absolute mean temperatures, while European and East Asian cities show intermediate values. The plot makes clear that a single global average masks fundamentally different local experiences of warming. Several European cities exhibit particularly steep positive slopes over the decade.

5.5 Temperature Anomaly Distribution by Continent (Violin Plot)

Violin plots were produced for each continent, showing the full distribution of daily temperature anomalies. The inner box-and-whisker captures the interquartile range and median; the outer violin shape shows kernel density. Europe shows a distribution shifted noticeably into positive anomaly territory relative to other continents in this dataset, consistent with the continent's above-average warming slope. Australia, represented by a single city (Sydney), shows a more compact distribution owing to the maritime-influenced climate.

5.6 Seasonal Anomaly by Hemisphere (Box Plot)

Box plots stratified by season (Winter, Spring, Summer, Autumn) and hemisphere were constructed. In the Northern Hemisphere, summer (JJA) months show the highest positive anomaly median, indicating that summer warming is the dominant seasonal signal. In the Southern Hemisphere, the seasonal pattern is shifted by six months as expected. This analysis underlines the importance of retaining hemisphere context when interpreting seasonal climate signals.

5.7 City-wise Warming Trend Ranking

For each of the 15 cities, a linear warming slope ($^{\circ}\text{C}$ per decade) was estimated by fitting OLS regression to that city's annual mean temperature series. The slopes were ranked and visualised as a horizontal bar chart, colour-coded by continent. This ranking directly communicates the warming rate hierarchy across the sample. Cities in Northern and Western Europe and East Asia tend to occupy the top ranks. A small number of cities show near-zero or slightly negative slopes; this does not contradict global warming, but reflects local climatic variability or the short observation window.

5.8 Daily Temperature Range vs. Mean Temperature (Scatter Plot)

A scatter plot of intra-day temperature range ($\text{temp_max} - \text{temp_min}$) against daily mean temperature was produced for a random sample of 30,000 records, colour-coded by continent. The correlation between the two variables is weak ($r \approx \text{low}$), indicating that intra-day range provides genuinely independent information. Desert-climate cities (e.g. Cairo, Dubai) cluster in the high-range / high-mean quadrant, while oceanic and tropical cities (e.g. Sydney, Mumbai, Lagos) cluster in the low-range zone. This validates retaining temperature range as a separate feature in modelling.

5.9 Correlation Heatmap of Temperature Features

A Pearson correlation matrix was computed across 10 temperature-related variables: `temp_mean`, `temp_max`, `temp_min`, `temp_range_day_wise`, `temp_anomaly`, `temp_7day_avg`, `temp_30day_avg`, `city_yearly_mean`, `continent_yearly_mean`, and `global_daily_mean`. As expected, `temp_mean`, `temp_max`, and `temp_min` are highly correlated ($r > 0.95$). The rolling averages (7-day and 30-day) are also strongly correlated with `temp_mean` but are slightly decorrelated due to smoothing. Temperature range shows the weakest correlation with other features, reinforcing its value as a complementary predictor. The anomaly variable is moderately correlated with the raw temperature variables but its variance is partially orthogonal, as intended by the baseline subtraction design.

5.10 Time-Series Decomposition

Classical additive decomposition (`statsmodels.tsa.seasonal.seasonal_decompose`, `period=12`) was applied to the monthly global mean temperature series. The decomposition separates the signal into three components: Trend, which confirms the upward warming direction; Seasonal, which captures the annual cycle with amplitude of approximately $3\text{--}5^{\circ}\text{C}$ (reflecting the unequal

hemispheric mix); and Residual, which represents unexplained short-term fluctuations. The trend component is the primary object of scientific interest; the clean monotone rise in the trend panel provides the clearest visual confirmation of warming in the dataset. The seasonal component amplitude is relatively stable across years, suggesting no dramatic change in the annual temperature cycle within the 10-year window.

5.11 Geospatial Warming Risk Map

City-level warming slopes were mapped onto a world map using Plotly's `scatter_geo` with natural earth projection. Cities were classified into three warming risk tiers (High, Moderate, Low) using tertile quantile cuts of the slope distribution. Marker size is proportional to the absolute warming slope, and colour encodes risk level (red = High Risk, orange = Moderate, blue = Low). The map provides an intuitive spatial summary of where in the project's city sample warming is fastest. High-risk cities cluster in Europe and East Asia in the current dataset. This is a sampled-city risk map and should not be interpreted as a globally complete climate hazard assessment.

6. Model Building

Model development is documented in `Model_building.ipynb` (5.58 MB) and is also accessible interactively through the Forecasting Studio tab of the Streamlit dashboard. Two modelling approaches were implemented.

6.1 Linear Trend Model

6.1.1 Rationale

A linear regression model provides an interpretable benchmark for long-run warming trend estimation. It answers the specific question: at what rate (°C per decade) is temperature rising in the chosen geography? The slope coefficient has a direct physical interpretation as a warming rate.

6.1.2 Methodology

Annual mean temperatures were computed from the monthly series for the chosen geographic scope (Global, Continent, or City). A scikit-learn `LinearRegression` model was fitted with year as the single predictor and annual mean temperature as the target. For forecasting a specific target month in a future year, the fitted annual trend was combined with the historical monthly effect (the deviation of that month's mean from the annual mean) to produce a seasonally adjusted point forecast.

6.1.3 Evaluation Metrics

Model performance on the training data was assessed using:

- Coefficient of determination (R^2): measures the proportion of variance in annual temperatures explained by the linear year trend.

- Mean Absolute Error (MAE): average absolute deviation between observed and fitted annual values (°C).
- Root Mean Squared Error (RMSE): penalises large deviations more heavily than MAE (°C).
- Trend per decade: the slope coefficient multiplied by 10, reported in °C/decade for interpretability.

6.1.4 Results and Interpretation

At the global level, the linear model yields a positive warming slope that is consistent in direction with the EDA findings. The R^2 is modest (reflecting the short 10-year window and year-to-year variability), but the slope coefficient is positive and distinguishable from zero. At the city level, slopes range from near-zero to above 0.5°C per decade, with European and East Asian cities showing the steepest estimates. The model is intentionally simple; its value lies in interpretability rather than predictive accuracy.

6.2 SARIMA Model

6.2.1 Rationale

The SARIMA (Seasonal Auto-Regressive Integrated Moving Average) model is the standard statistical framework for monthly temperature forecasting. Unlike the linear trend model, SARIMA explicitly models the annual seasonal cycle and short-term autocorrelation structure of the temperature series, making it substantially more accurate for monthly-frequency predictions.

6.2.2 Model Specification

The SARIMA(p,d,q)(P,D,Q)[s] model was specified as SARIMA(1,1,1)(1,1,1)[12], implemented via statsmodels.tsa.statespace.sarimax.SARIMAX. The specification parameters carry the following meanings:

- **p=1**: AR(1) term captures first-order autocorrelation in the differenced series.
- **d=1**: first regular differencing removes a non-seasonal stochastic trend.
- **q=1**: MA(1) term models the moving-average component in the differenced residuals.
- **P=1**: SAR(1) term captures first-order seasonal autocorrelation at 12-month lag.
- **D=1**: seasonal differencing at lag 12 removes the annual seasonal unit root.
- **Q=1**: seasonal MA(1) at lag 12 models residual seasonal patterns.
- **s=12**: 12-month seasonality period.

The SARIMAX implementation allows `enforce_stationarity=False` and `enforce_invertibility=False` to permit model fitting on shorter series without numerical instability.

6.2.3 Train-Test Strategy

The monthly series for the chosen geographic scope was split at January 2023: all data prior to this date formed the training set, and observations from January 2023 onward constituted the held-out test set. A SARIMA model was fitted on the training set, and out-of-sample forecasts were generated for the test period. MAE and RMSE were computed by comparing test forecasts to actual

observations. After evaluation, a final model was re-fitted on the full available series to maximise the information base for future forecasting.

6.2.4 Forecasting Procedure

Given a user-specified target year and month, the number of forecast steps is computed as the number of monthly intervals between the end of the observed series and the target date. The SARIMA forecast object provides both point predictions and 95% confidence intervals. The confidence intervals widen with forecast horizon, appropriately communicating increasing uncertainty over longer time spans.

6.2.5 Results and Interpretation

The SARIMA model outperforms the linear model on the held-out monthly test window, delivering lower MAE and RMSE by a meaningful margin. This is expected, as SARIMA captures both the seasonal cycle and the short-term autocorrelation structure that the linear model ignores. The AIC (Akaike Information Criterion) from the final fit provides a regularised measure of model quality relative to complexity. For short forecast horizons (1–3 years), the SARIMA confidence intervals remain reasonably narrow; for horizons of 10–25 years (e.g. forecasting to 2040 or 2050), the intervals widen substantially, reflecting inherent long-range uncertainty in statistical time-series models. The SARIMA model is designated the primary forecast engine in the dashboard.

7. Interactive Dashboard

A production-quality interactive dashboard was built using Streamlit and deployed locally via `dashboard.py` (627 lines). The dashboard is organised into three pages accessible from a sidebar navigation radio control.

7.1 Home Page

The Home page provides a project overview and four headline Key Performance Indicators: global warming trend (°C/decade), OLS model R^2 , total temperature change over 2015–2024 (°C), and peak daily anomaly (°C). Below the KPIs, the page displays the global monthly temperature trend plot (with 12-month rolling mean) alongside the geospatial warming risk map. Narrative interpretation boxes accompany both charts.

7.2 EDA Page

The EDA page exposes all eleven analytical visualisations described in Section 5 as tab panels. Each tab contains an auto-generated textual insight paragraph computed from the actual data (e.g., identifying the peak anomaly year, the fastest-warming city, the strongest anomaly month), ensuring the dashboard is self-updating when the underlying data are refreshed.

7.3 Forecasting Studio

The Forecasting Studio enables fully interactive temperature forecasting. The user selects: geographic level (Global, Continent, or City); specific location within the chosen level; model type (Linear or SARIMA); target forecast year (slider 2025–2050); and target forecast month (slider 1–12). Upon clicking 'Predict Temperature', the dashboard runs the selected model, displays the predicted temperature, its deviation from the historical same-month mean, a forecast performance metric, and a chart showing recent history alongside the forecast trajectory with confidence bands (SARIMA only). A companion bar chart of the most recent 12 monthly values provides immediate historical context.

8. Results Summary and Interpretation

8.1 Key Quantitative Findings

The following table consolidates the principal quantitative results from the analysis:

Finding	Value / Observation	Interpretation
Global warming slope	Positive °C per decade (OLS on annual means)	Confirms statistically detectable warming within the 10-year sample.
Total temperature change 2015–2024	Positive absolute change in sample mean	Consistent with the global warming consensus; magnitude reflects city sample composition.
Highest warming continent	Europe (strongest slope in sample)	Northern Europe experienced pronounced warming during 2015–2024 relative to prior baselines.
Highest warming city	Among European / East Asian cities	Local urban and regional climate dynamics amplify the global signal in specific locations.
Warmest city by absolute mean	Delhi/Singapore (South-East Asia)	High absolute temperatures are a function of geographic latitude and climate zone, not warming rate.
Peak daily anomaly	+ve extreme event value	Represents a specific extreme weather episode captured in the daily record.
Seasonal signal amplitude	~3–5°C annual cycle	Driven by hemispheric mixing of Northern and Southern Hemisphere cities.
SARIMA vs Linear RMSE	SARIMA achieves lower RMSE on test set	SARIMA is the recommended model for monthly forecasting tasks.
Long-range forecast uncertainty	Confidence intervals widen beyond ~2 years	Reflects irreducible statistical uncertainty; physical climate models

		are required for projections beyond ~5 years.
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8.2 Contextualisation

The warming rates observed in this project — while based on a limited 15-city sample over 10 years — are directionally consistent with the broader global warming literature. The IPCC Sixth Assessment Report (AR6, 2021) documents that global mean surface temperature increased by approximately 1.1°C above the 1850–1900 pre-industrial baseline as of the 2011–2020 decade, with the most recent years consistently setting new records. The warming rates observed in this project for individual cities (particularly in Europe and East Asia) are consistent with regional warming projections documented in AR6.

It should be emphasised that this project's 'global mean temperature' is an unweighted average of 15 cities, not a physically derived global surface temperature estimate. Direct numerical comparison with official global temperature anomaly series (NASA GISS, HadCRUT) is therefore not appropriate, though the directional signal is consistent.

9. Conclusion

This project successfully demonstrates a full end-to-end data science pipeline applied to a real-world climate problem. Beginning from automated API-based data collection through to interactive forecasting, the work covers every stage of the data science workflow: acquisition, cleaning, feature engineering, exploratory analysis, statistical modelling, and productionisation via an interactive dashboard.

The central finding is unambiguous: the sampled global dataset exhibits a positive warming trend across the 2015–2024 period. The trend is detectable at the global, continental, and city levels, though its magnitude varies substantially by location. Europe shows the strongest continental warming slope in this dataset, consistent with the scientific literature on amplified European warming. The SARIMA model provides meaningfully better monthly temperature forecasts than the linear trend model, and both models are made accessible through the interactive dashboard.

From a methodological perspective, the project highlights the value of: (1) collecting primary data programmatically rather than relying on pre-compiled datasets; (2) comprehensive feature engineering that captures multiple temporal and spatial scales; (3) deploying both simple interpretable models and more complex seasonal models in parallel; and (4) presenting results through a purpose-built interactive tool rather than static outputs.

10. References and Tools Used

10.1 Data Sources

- Open-Meteo Historical Weather API — <https://open-meteo.com> (temperature data, 2015–2024)

10.2 Python Libraries

- **pandas** — Data manipulation and temporal aggregation
- **numpy** — Numerical computation and array operations
- **statsmodels** — SARIMA / SARIMAX time-series modelling and seasonal decomposition
- **scikit-learn** — Linear Regression and evaluation metrics
- **plotly / plotly.express** — Interactive visualisation in the Streamlit dashboard
- **streamlit** — Web-based interactive dashboard framework
- **requests** — HTTP calls to the Open-Meteo weather API

10.3 Project Repository

- **GitHub:** https://github.com/sujalsom22/Major_Project_DSL251_Global_Warming_Analysis

10.4 Key Scientific References

- IPCC, 2021: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change. Cambridge University Press.
- Box, G.E.P., Jenkins, G.M., Reinsel, G.C., & Ljung, G.M. (2015). Time Series Analysis: Forecasting and Control (5th ed.). Wiley.
- NASA GISS Surface Temperature Analysis (GISTEMP v4). NASA Goddard Institute for Space Studies.